Chapter 8 — Vertex Colourings

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1 Chromatic Number

1.1 What is a vertex colouring?

Let $G = (V, E)$ be a simple graph (no loops, no multiple edges).
A vertex colouring of G is a function:

$$c : V \to \{1, 2, \ldots, k\}$$

such that:

$$\text{if } uv \in E, \quad c(u) \neq c(v)$$

Interpretation

- Each vertex is assigned a colour (represented by an integer).
- Adjacent vertices must receive different colours.
- Non-adjacent vertices may share the same colour.
- This is a constraint satisfaction problem imposed by adjacency.

1.2 Proper colourings and k-colourability

A colouring that satisfies the adjacency condition is called a **proper colouring**.

Definition A graph G is **k-colourable** if there exists a proper vertex colouring using at most k colours.

Key points:

- "At most" is important.
- If a graph is k-colourable, it is also $(k+1)$-colourable, $(k+2)$-colourable, etc.
- The interesting question is the smallest such k.

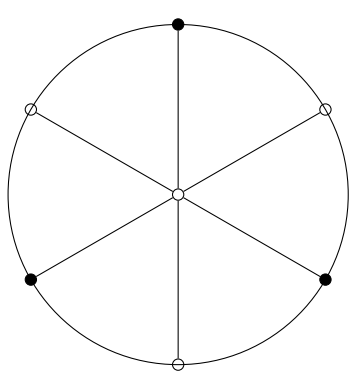

Figure 1: A 3-chromatic graph.

1.3 The chromatic number

Definition (central concept) The chromatic number of a graph G, denoted $\chi(G)$, is the minimum number of colours required for a proper vertex colouring of G.

$$\chi(G) = \min\{k \mid G \text{ is } k\text{-colourable}\}$$

This number measures the intrinsic colouring complexity of the graph.

1.3.1 First examples (essential intuition)

Empty graph E
Let E_n be a graph with n vertices and no edges.

- No adjacency constraints.
- All vertices may receive the same colour.
- $\chi(E_n) = 1 \quad (n \geq 1)$

Complete graph K_n
Every pair of vertices is adjacent.

- Each vertex must have a different colour.
- $\chi(K_n) = n$
- This is the maximum possible chromatic number for a graph with n vertices.

Paths P_n
A path alternates adjacency.

- Colour vertices alternately (e.g., red–blue–red–blue).
- Works for any length.
- $\chi(P_n) = 2 \quad (n \geq 2)$

Cycles C_n

- If n is even: $\chi(C_n) = 2$
- If n is odd: $\chi(C_n) = 3$
- The odd cycle is the smallest obstruction to 2-colourability.

1.4 Bipartite graphs and 2-colourability

Definition A graph is bipartite if its vertex set can be partitioned into two sets $V = A \cup B$ such that no edge joins two vertices in the same set.

Fundamental theorem A graph is bipartite if and only if $\chi(G) \leq 2$.
Equivalently: A graph is 2-colourable $\iff$ It contains no odd cycle.

1.5 Lower bounds for the chromatic number

Determining $\chi(G)$ exactly is hard, so we study bounds.

Clique number A clique is a complete subgraph. Let $\omega(G) =$ size of the largest clique in G. Since all vertices in a clique are pairwise adjacent:

$$\chi(G) \geq \omega(G)$$

Maximum degree Let $\Delta(G) = \max_{v \in V} \deg(v)$.
Trivially:

$$\chi(G) \leq \Delta(G) + 1$$

(Reason: colour greedily.)

1.6 The greedy colouring algorithm

Order the vertices arbitrarily: $v_1, v_2, \ldots, v_n$. Colour each vertex with the smallest available colour not used by its coloured neighbours.

Properties

- Uses at most $\Delta(G) + 1$ colours.
- Fast and simple but not optimal in general.
- Strongly dependent on vertex ordering.

1.7 Chromatic number vs intuition

Important warning: Sparse graphs can have large chromatic numbers. There exist graphs with no triangles, large girth, and arbitrarily large chromatic numbers. Chromatic number is not controlled by local density alone; global structure matters.

1.8 Computational complexity

Determining whether $\chi(G) \leq k$ for a given $k \geq 3$ is **NP-complete**.

- No known efficient algorithm.
- Exact computation is infeasible for large graphs.

2 Brooks' Theorem

2.1 Motivation: why Brooks' Theorem matters

From Section 8.1 we know:

$$\chi(G) \leq \Delta(G) + 1$$

This bound is always true, but often far from tight.

Natural question When do we actually need $\Delta(G) + 1$ colours? Brooks' Theorem answers this almost completely.

2.2 Statement of Brooks' Theorem

Theorem (Brooks, 1941)
Let G be a connected graph. Then:

$$\chi(G) \leq \Delta(G)$$

unless G is:

1. A complete graph, or
2. An odd cycle.

Equivalently:

$$\chi(G) = \begin{cases} \Delta(G) + 1 & \text{if } G = K_n \text{ or } G = C_{2k+1}, \\ \leq \Delta(G) & \text{otherwise.} \end{cases}$$

This result is best possible.

2.2.1 Understanding the exceptions

Complete graphs For K_n:

- $\Delta(K_n) = n - 1$
- $\chi(K_n) = n = \Delta + 1$

This is unavoidable because every vertex is adjacent to all others.

Odd cycles For C_{2k+1}:

- $\Delta = 2$
- $\chi = 3 = \Delta + 1$

As seen earlier, odd cycles cannot be 2-coloured.

Key insight These are the only connected graphs that force the extra colour. Everything else can be coloured with at most $\Delta(G)$ colours.

2.2.2 Why the theorem is surprising

Intuitively, one might think high-degree vertices or local congestion force many colours. Brooks' Theorem shows:

- Degree alone *almost* determines $\chi(G)$.
- Except for two very specific, rigid structures, $\Delta(G)$ is always enough.

2.2.3 Examples to internalize the theorem

Even cycle C_6
$\Delta = 2$. Not complete and not odd $\Rightarrow \chi(C_6) \leq 2$. Indeed, $\chi(C_6) = 2$.
A cubic graph (3-regular)
Let G be connected and every vertex has degree 3 ($\Delta = 3$). If $G \neq K_4$, then $\chi(G) \leq 3$. Many nontrivial graphs (e.g., the Petersen graph) satisfy this.
Trees
For all trees except K_1, $\Delta \geq 1$. Since trees are bipartite, $\chi(G) = 2 \leq \Delta$ (provided $\Delta \geq 2$; if $\Delta = 1$, the tree is K_2, which is an exception covered by the K_n rule).

2.2.4 Proof idea (high-level)

The full proof is nontrivial and constructive. At an intuitive level:

- Choose a vertex ordering that avoids conflicts.
- Exploit the absence of cliques or odd cycles.
- Use Kempe chains (recolouring arguments) to resolve bottlenecks.
- Force a greedy colouring to succeed with Δ colours.

2.2.5 Why connectivity matters

Brooks' Theorem assumes connected graphs. If G is disconnected, apply the theorem to each component and take the maximum $\chi(G)$ among them.

2.2.6 Relation to earlier bounds

Recall: $\omega(G) \leq \chi(G) \leq \Delta(G) + 1$.
Brooks refines this to:

$$\omega(G) \leq \chi(G) \leq \Delta(G)$$

for all but two exceptional families. This is a dramatic tightening of the upper bound.

2.2.7 Consequences and applications

- Shows that high chromatic number requires global structure.
- Motivates the study of critical graphs and the relationship between girth and chromatic number.

2.2.8 Conceptual Interpretation

Brooks' theorem characterizes graphs that are *degree-critical* with respect to coloring. It plays a central role in the theory of critical graphs and serves as a foundational result in structural graph theory.

2.3 Grötzsch Graph

The Grötzsch Graph. The Grötzsch graph is the smallest triangle-free planar graph with chromatic number 4. It has 11 vertices and 20 edges. Its importance lies in showing that **even planar graphs with no triangles can require four colors**, demonstrating that forbidding short cycles alone does not reduce the chromatic number below 4.

Remark 2.1. The Grötzsch graph connects directly to the theorem of the existence of triangle-free graphs with arbitrary chromatic number, providing a concrete, planar, minimal example for $k = 4$. It is an extremal example in planar graph coloring, illustrating that girth constraints do not automatically lower $\chi(G)$ in planar graphs.

2.4 The Four-Color Theorem.

Theorem 2.1 (Four-Color Theorem)**.** Every planar graph can be properly colored with at most four colors, i.e., if G is planar, $\chi(G) \leq 4$.

Remark 2.2. - It implies that no planar graph requires more than four storage units in the planar storage analogy. - While the theorem sets a universal upper bound for planar graphs, the Grötzsch graph shows that even triangle-free planar graphs may achieve the maximum $\chi(G) = 4$, so local sparsity is not sufficient to guarantee a 3-coloring. - Historically, this theorem was the first major result proven with extensive computer assistance (Appel and Haken, 1976), demonstrating both the complexity and subtlety of planar graph coloring.

Connection to previous results: - Brooks' theorem: In planar graphs, $\Delta \leq 5$, so Brooks' theorem guarantees $\chi(G) \leq \Delta(G)$ except for exceptional cases; the Four-Color Theorem provides a stronger global bound for planarity. - Chromatic polynomials: For planar graphs, $P_G(4) > 0$ always; Grötzsch graph shows $P_G(3) = 0$ even though the graph is triangle-free. - Girth vs. chromatic number: The Grötzsch graph is triangle-free ($girth \geq 4$) but still requires four colors.

Illustrative Example: - Grötzsch graph (11 vertices) requires 4 colors; every proper 3-coloring attempt fails due to forced color conflicts, showing explicitly that $\chi(G) = 4$ for a planar triangle-free graph.

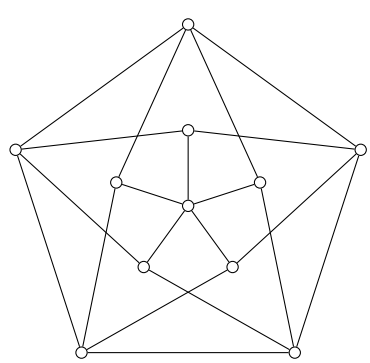

Figure 2: The Grötzsch graph—a 4-critical graph.

3 Hajós' Conjecture

3.1 Motivation: beyond Brooks' Theorem

From Brooks' Theorem we learned that large degree alone does not force a large chromatic number. Only very rigid structures (cliques, odd cycles) force extremal behaviour.

Natural question If a graph has a large chromatic number, what must it look like? Hajós' Conjecture is an attempt to answer exactly this by turning colouring theory into structural graph theory.

3.2 Hajós' Conjecture — informal idea

Very loosely: Graphs with large chromatic number must contain "highly connected" structures resembling complete graphs.

3.3 Precise statement of Hajós' Conjecture

Conjecture (Hajós, 1940s)
If a graph G satisfies $\chi(G) \geq k$, then G contains a **subdivision** of K_k.

3.4 What does "subdivision of K_k" mean?

This is the critical technical notion, also known as a **topological minor**.

Subdivision A subdivision of a graph H is obtained by replacing edges of H by paths, where these paths are internally vertex-disjoint.

- Vertices of H (branch vertices) remain.
- Edges may be "stretched" into paths.

Examples

- **Subdivision of K_3:** K_3 is a triangle. A subdivision replaces each edge with a path (e.g., a cycle C_n for $n > 3$). It still "looks like" a triangle topologically.
- **Subdivision of K_4:** Contains four branch vertices where every pair is joined by internally disjoint paths.

3.5 Why Hajós' Conjecture is plausible

- K_k itself has chromatic number k.
- Subdividing edges does not reduce the chromatic number (it usually maintains it or makes it harder to colour).
- High chromatic number suggests a dense interconnection, or a "hidden K_k inside."

3.5.1 Known results: what is true and what is false

True for small k

- $k = 1, 2, 3$: Trivial.
- $k = 4$: True.
- $k = 5$: True (requires a non-trivial proof).

False in general For $k \geq 6$, **Hajós' Conjecture is false**. There exist graphs with arbitrarily large chromatic numbers that contain no subdivision of K_k. This counter-intuitive result was a major surprise in the field.

3.5.2 What replaces Hajós' Conjecture?

Although false, it inspired several deep results regarding the structure of graphs.

k-critical graphs A graph G is **k-critical** if:

- $\chi(G) = k$
- Removing any edge or vertex reduces the chromatic number.

These are the minimal obstructions to $(k-1)$-colourability.

3.6 Connection to girth and extremal examples

Erdős showed (probabilistically) that graphs can have **arbitrarily large girth** (no short cycles) and **arbitrarily large chromatic number**. These graphs have no large cliques, yet still require many colours, breaking naive intuition.

3.6.1 Conceptual importance

Hajós' Conjecture framed the right questions, connecting colouring to topological structure and influencing:

- Minors theory.
- Hadwiger's Conjecture (a related, still open conjecture regarding graph minors).
- Modern structural graph theory.

4 Chromatic Polynomials

4.1 Motivation: from "how many colours?" to "how many colourings?"

Up to now, we asked questions like: "Can G be coloured with k colours?" or "What is $\chi(G)$?" Now we ask: *Given k colours, how many proper colourings does G have?* Surprisingly, the answer is always a polynomial in k.

4.2 Definition of the chromatic polynomial

Definition For a graph G, the chromatic polynomial $P_G(k)$ is defined as the number of proper vertex colourings of G using k colours.

- k is a variable (usually a positive integer).
- $P_G(k)$ counts proper colourings.
- Colours are labelled (red, blue, green, etc.).

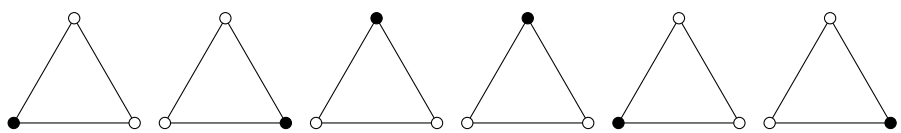

Figure 3: Placeholder

A fundamental theorem due to Birkhoff and Whitney asserts that $P_G(k)$ is indeed a polynomial in k.

4.3 Fundamental theorem

Theorem: For every finite graph G, $P_G(k)$ is a polynomial in k of degree $|V(G)|$.

4.4 Immediate consequences

Degree If $|V(G)| = n$, then $\deg P_G(k) = n$. The leading term is always k^n. *Reason:* With no constraints (no edges), each vertex can be coloured independently in k ways.

Integer coefficients with alternating signs The polynomial has the form:

$$P_G(k) = k^n - a_1 k^{n-1} + a_2 k^{n-2} - \cdots$$

with $a_i \geq 0$. This reflects inclusion–exclusion over adjacency constraints.

4.5 First concrete examples

Empty graph E_n
No edges $\Rightarrow$ no constraints. $P_{E_n}(k) = k^n$.

Complete graph K_n
Every vertex must have a distinct colour.

$$P_{K_n}(k) = k(k-1)(k-2)\cdots(k-n+1)$$

This is a falling factorial, often denoted $(k)_n$.

Path P_n

- First vertex: k choices.
- Each subsequent vertex: $k-1$ choices.
- $P_{P_n}(k) = k(k-1)^{n-1}$.

Cycle C_n

$$P_{C_n}(k) = (k-1)^n + (-1)^n(k-1)$$

This formula explains why even cycles are 2-colourable and odd cycles are not.

4.6 Deletion–contraction recurrence

Theorem (Deletion–Contraction) Let e be an edge of G. Then:

$$P_G(k) = P_{G-e}(k) - P_{G/e}(k)$$

Where:

- $G - e$: graph with edge e deleted.
- G/e: graph with edge e contracted (endpoints merged).

Intuition $P_{G-e}(k)$ counts colourings ignoring the constraint at e, while $P_{G/e}(k)$ counts colourings where the endpoints of e get the same colour. Subtracting the latter enforces the "different colours" rule.

4.6.1 Example: $C_3 = K_3$

Pick an edge e.

- $G - e = P_3$
- $G/e = K_2$

$P_{C_3}(k) = P_{P_3}(k) - P_{K_2}(k) = k(k-1)^2 - k(k-1) = k(k-1)(k-2)$.

4.7 Chromatic number from the polynomial

$$\chi(G) = \min\{k \in \mathbb{N} \mid P_G(k) > 0\}$$

The polynomial encodes the chromatic number, but contains much more information about the graph's structure.

4.8 Zeros and Importance

- Values where $P_G(k) = 0$ are **chromatic roots**. All non-negative integers less than $\chi(G)$ are roots.
- They connect graph theory to statistical physics (Potts model) and network reliability.

5 Girth and Chromatic Number

5.1 The naïve intuition (that turns out to be wrong)

From earlier sections, it is natural to believe that short cycles (especially triangles and odd cycles) force many colours, or that a large chromatic number must come from local density. One might assume that if a graph has no short cycles, it should be "easy" to colour. This intuition is false.

The girth of a graph G, denoted $g(G)$, is the length of the shortest cycle in G.

- **Trees:** $g(G) = \infty$.
- **Triangle-free graphs:** $g(G) \geq 4$.
- **Bipartite graphs:** Contain no odd cycles, but may still have short even cycles (e.g., C_4).

5.2 The central question

How are girth and chromatic number related? Specifically:

- Does large girth force a small chromatic number?
- Does a small chromatic number imply short cycles?

We know odd cycles force $\chi \geq 3$, but higher values of χ are more mysterious.

The shocking result (Erdős, 1959)

Theorem (Erdős)
For every pair of integers $g \geq 3$ and $k \geq 2$, there exists a graph G such that:

$$g(G) \geq g \quad \text{and} \quad \chi(G) \geq k$$

Interpretation You can simultaneously have no short cycles (arbitrarily large girth) and a very large chromatic number. This shows that a large chromatic number does **not** require local density or short cycles; it destroys the "local obstruction" viewpoint.

5.2.1 Why this is counter-intuitive

Recall that bipartite graphs have no odd cycles and $\chi \leq 2$. Odd cycles were the only obstruction to 2-colourability. One might expect higher chromatic numbers to require higher-order local structures, but Erdős proved that colouring difficulty can be globally distributed. No small local pattern explains it.

5.2.2 Proof idea (The Probabilistic Method)

The proof uses the probabilistic method, which was revolutionary in combinatorics.

High-level strategy

1. Start with a random graph on n vertices.
2. Show that, with positive probability, the graph has very few short cycles.
3. Show that, with positive probability, it has a small **independence number** $\alpha(G)$.
4. Remove one vertex from every short cycle.

The Resulting Graph The final graph has a large girth (because we broke all short cycles) and a small maximum independent set.

Key Insight Recall the relationship:

$$\chi(G) \geq \frac{|V(G)|}{\alpha(G)}$$

By keeping $\alpha(G)$ very small relative to the number of vertices, we force the chromatic number to be arbitrarily large, even without any local cliques or triangles.

5.3 The Mycielski Construction

5.3.1 Why Mycielski's construction is important

While Erdős' theorem (Section 8.5) is probabilistic and non-constructive, the Mycielski construction is explicit, deterministic, and elementary. It provides concrete examples showing that one can increase the chromatic number without increasing the clique number, directly complementing Erdős' results.

5.3.2 The problem Mycielski addressed

We know $\chi(G) \geq \omega(G)$ (cliques force colours). The question Mycielski answered was: *Can we force a high chromatic number while keeping the clique number small?* His answer was a definitive yes, even for triangle-free graphs ($\omega = 2$).

5.3.3 Definition

Let $G = (V, E)$ be a graph with $V = \{v_1, v_2, \ldots, v_n\}$. The Mycielskian $\mu(G)$ is constructed as follows:

1. **Step 1: Duplicate vertices.** Create a new set $U = \{u_1, u_2, \ldots, u_n\}$ where each u_i corresponds to v_i.
2. **Step 2: Copy adjacency.** For every edge $v_iv_j \in E$, add "cross edges" u_iv_j and u_jv_i. (Note: no edges are added between u_i and u_j).
3. **Step 3: Add a new vertex.** Add a final vertex w and connect it to every vertex in U.

Summary of $\mu(G)$:

- $V(\mu(G)) = V \cup U \cup \{w\}$.
- Edges include the original edges in V, the cross edges between U and V, and the star edges from w to U.

5.3.4 Key properties

Theorem 1: $\chi(\mu(G)) = \chi(G) + 1$.
The vertex w forces an extra colour, and a clever recolouring argument shows exactly one more is sufficient.

Theorem 2: $\omega(\mu(G)) = \omega(G)$.
In particular, if G is triangle-free, $\mu(G)$ remains triangle-free.

Theorem 3: If G has girth ≥ 4, then $\mu(G)$ has girth ≥ 4.

5.3.5 Concrete example

Start with the cycle $G = C_5$:

- $\chi(C_5) = 3$ and it is triangle-free.
- $\mu(C_5)$ (the **Grötzsch graph**) has $\chi = 4$ and is still triangle-free.
- Applying it again, $\mu(\mu(C_5))$ has $\chi = 5$ and remains triangle-free.

This allows for the creation of an infinite family of triangle-free graphs with unbounded chromatic number.

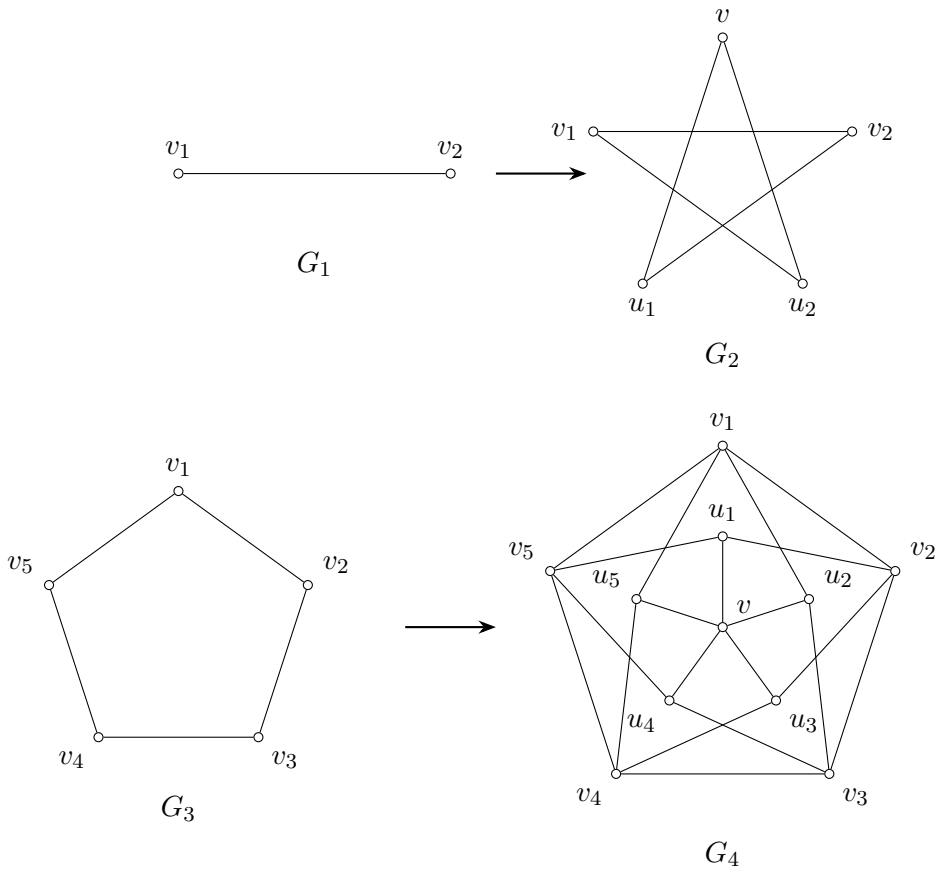

Figure 4: Mycielski's Construction

5.3.6 Conceptual meaning

The Mycielski construction shows that colouring difficulty can be forced globally in a controlled way. It provides a constructive counterweight to Erdős' probabilistic method and explains why local density (cliques) is not the sole driver of the chromatic number.

Relationship to Hajós' Conjecture

Mycielski graphs are useful counter-examples in colouring theory because they possess a high chromatic number without containing large complete subgraphs, illustrating why conjectures like Hajós' (which look for K_k subdivisions) become so difficult to satisfy as k increases.

6 A Storage Problem

6.1 The real-world problem (informal statement)

Consider a collection of items (chemicals, data blocks, or tasks) where certain pairs cannot be stored together due to interference, incompatibility, or resource conflicts. The goal is to:

- Store all items using the **minimum number of storage units**.
- Ensure no incompatible items share the same unit.

This is a classic optimization problem.

6.2 Graph-theoretic modelling

We translate this real-world scenario into the language of graph theory:

Step 1: Construct the graph Define a graph $G = (V, E)$ as follows:

- **Vertices (V):** Each item corresponds to a vertex.
- **Edges (E):** An edge uv exists if items u and v are incompatible.

This is known as a **conflict graph**.

Step 2: Interpret storage units as colours

- Assign a colour to each vertex.
- Vertices of the same colour represent items stored together in the same unit.
- Adjacent vertices must have different colours to avoid conflicts.

This is exactly a **proper vertex colouring**.

6.3 Core equivalence

The minimum number of storage units required is the **chromatic number** $\chi(G)$ of the conflict graph.

Why this formulation is powerful Once modelled, all results from Sections 8.1–8.5 apply:

- **Brooks' Theorem:** Provides degree-based bounds for storage units.
- **Chromatic Polynomials:** Allows us to count the number of ways to assign items to k units.
- **Girth/Mycielski:** Warns us that global complexity can remain high even if items only conflict with a few neighbors.

Example: Chemical Storage Suppose 5 chemicals have incompatibilities forming a cycle C_5.

- Maximum degree $\Delta = 2$.
- However, $\chi(C_5) = 3$.

Conclusion: Even though each chemical conflicts with only two others, three storage units are unavoidable. This mirrors Brooks' Theorem (odd cycles).

6.4 Why greedy strategies may fail

A naïve "first-available-spot" strategy is essentially a **greedy colouring**.

- **Problem:** Depending on the order of items, the algorithm may use more than $\chi(G)$ units.
- This reflects the **NP-hardness** of finding an optimal storage solution for large, complex conflict graphs.

6.5 Interpretation using independent sets

Each storage unit represents an **independent set** in G. The storage problem is fundamentally a task of partitioning $V(G)$ into the minimum number of independent sets. This connects graph colouring to:

- Scheduling (tasks using the same resource).
- Channel assignment in telecommunications.
- Register allocation in compiler design.